Congress Stock Market Trades 2012-2025

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Setting wd and reading excel file

```
setwd("~/Desktop/J_Coder/Data_334")

library(readxl)

# Attempt to read the file as an Excel file
cog <- read_excel("congress-trading-all.xlsx")
```

```
library(haven)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.1     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Str

```
str(cog)
```

```
## tibble [48,422 x 20] (S3: tbl_df/tbl/data.frame)
##  $ Ticker            : chr [1:48422] "SO" "PANW" "NEE" "TSCO" ...
##  $ TickerType        : chr [1:48422] "ST" "ST" "ST" "ST" ...
##  $ Company           : chr [1:48422] "SOUTHERN COMPANY" "PALO ALTO NETWORKS, INC." "NEXTERA ENERGY, I
##  $ Traded            : chr [1:48422] "2024-11-07" "2024-11-07" "2024-11-07" "2024-11-07" ...
##  $ Transaction       : chr [1:48422] "Purchase" "Purchase" "Purchase" "Purchase" ...
##  $ Trade_Size_USD    : chr [1:48422] "$1,001 - $15,000" "$1,001 - $15,000" "$1,001 - $15,000" "$1,00
##  $ Status            : chr [1:48422] "NEW" "NEW" "NEW" "NEW" ...
##  $ Subholding        : chr [1:48422] NA NA NA NA ...
##  $ Description       : chr [1:48422] NA NA NA NA ...
##  $ Name              : chr [1:48422] "Marjorie Taylor Mrs Greene" "Marjorie Taylor Mrs Greene" "Marjo
##  $ BioGuideID        : chr [1:48422] "G000596" "G000596" "G000596" "G000596" ...
##  $ Filed             : chr [1:48422] "2024-11-08" "2024-11-08" "2024-11-08" "2024-11-08" ...
##  $ Party             : chr [1:48422] "R" "R" "R" "R" ...
##  $ District          : chr [1:48422] "GA14" "GA14" "GA14" "GA14" ...
##  $ Chamber           : chr [1:48422] "House" "House" "House" "House" ...
##  $ Comments          : chr [1:48422] NA NA NA NA ...
##  $ Quiver_Upload_Time: chr [1:48422] "2024-11-11" "2024-11-11" "2024-11-11" "2024-11-11" ...
##  $ excess_return     : chr [1:48422] "1.08631900833784" "2.34404486278548" "1.20458685435411" "-3.19
##  $ State             : chr [1:48422] "Georgia" "Georgia" "Georgia" "Georgia" ...
##  $ last_modified     : chr [1:48422] "2024-11-11" "2024-11-11" "2024-11-11" "2024-11-11" ...
```

```
head(cog)
```

```
## # A tibble: 6 x 20
##   Ticker TickerType Company  Traded Transaction Trade_Size_USD Status Subholding
##   <chr>  <chr>      <chr>    <chr>  <chr>       <chr>          <chr>  <chr>
## 1 SO     ST         SOUTHER~ 2024-~ Purchase    $1,001 - $15,~ NEW    <NA>
## 2 PANW   ST         PALO AL~ 2024-~ Purchase    $1,001 - $15,~ NEW    <NA>
## 3 NEE    ST         NEXTERA~ 2024-~ Purchase    $1,001 - $15,~ NEW    <NA>
## 4 TSCO   ST         TRACTOR~ 2024-~ Purchase    $1,001 - $15,~ NEW    <NA>
## 5 GS     ST         GOLDMAN~ 2024-~ Purchase    $1,001 - $15,~ NEW    <NA>
## 6 FDX    ST         FEDEX C~ 2024-~ Purchase    $1,001 - $15,~ NEW    <NA>
## # i 12 more variables: Description <chr>, Name <chr>, BioGuideID <chr>,
## #   Filed <chr>, Party <chr>, District <chr>, Chamber <chr>, Comments <chr>,
## #   Quiver_Upload_Time <chr>, excess_return <chr>, State <chr>,
## #   last_modified <chr>
```

reading and changing all columns names to lower case

```
colnames(cog)
```

```
##  [1] "Ticker"             "TickerType"         "Company"
##  [4] "Traded"             "Transaction"        "Trade_Size_USD"
##  [7] "Status"             "Subholding"         "Description"
```

```
## [10] "Name"               "BioGuideID"         "Filed"
## [13] "Party"              "District"           "Chamber"
## [16] "Comments"           "Quiver_Upload_Time" "excess_return"
## [19] "State"              "last_modified"
```

```
colnames(cog) <- tolower(colnames(cog))

print(colnames(cog))
```

```
##  [1] "ticker"             "tickertype"         "company"
##  [4] "traded"             "transaction"        "trade_size_usd"
##  [7] "status"             "subholding"         "description"
## [10] "name"               "bioguideid"         "filed"
## [13] "party"              "district"           "chamber"
## [16] "comments"           "quiver_upload_time" "excess_return"
## [19] "state"              "last_modified"
```

Independent variable, Party affiliation

```
table(cog$party)
```

```
##
##     D     I     R
## 22844    65 25513
```

```
table(cog$state)
```

```
##
##              Alabama               Alaska              Arizona
##                 1532                  118                  388
##             Arkansas           California             Colorado
##                  124                 3651                  268
##          Connecticut             Delaware District of Columbia
##                  147                  787                    5
##              Florida              Georgia               Hawaii
##                 4150                 3060                    8
##                Idaho              Illinois              Indiana
##                  140                  660                  122
##                 Iowa               Kansas             Kentucky
##                  222                  831                  554
##            Louisiana                Maine             Maryland
##                  184                  392                   17
##        Massachusetts             Michigan            Minnesota
##                  768                 1216                  872
##          Mississippi             Missouri              Montana
##                  589                    7                 1331
##             Nebraska               Nevada           New Jersey
##                   60                 1357                 6514
##           New Mexico             New York       North Carolina
##                    3                 2549                 1908
```

```
##         North Dakota                 Ohio             Oklahoma
##                  185                  520                 1048
##               Oregon         Pennsylvania         Rhode Island
##                  621                  532                 1360
##       South Carolina         South Dakota            Tennessee
##                   83                    1                 1698
##                Texas                 Utah              Vermont
##                 4406                  435                  102
##             Virginia           Washington        West Virginia
##                 1176                  576                 1095
##            Wisconsin              Wyoming
##                   47                    3
```

dependent variable, excess_return

```
cog$excess_return <- as.numeric(cog$excess_return)

# Remove rows with NA excess_return
cog_clean <- cog[!is.na(cog$excess_return), ]

pivot_view <- cog_clean[, c("name", "ticker", "traded", "trade_size_usd", "excess_return")]


pivot_view <- pivot_view[order(-pivot_view$excess_return), ]


head(pivot_view, 10)
```

```
## # A tibble: 10 x 5
##    name                    ticker traded     trade_size_usd   excess_return
##    <chr>                   <chr>  <chr>      <chr>                    <dbl>
##  1 Whitehouse, Sheldon     NVDA   2016-01-06 $1,001 - $15,000        18125.
##  2 Whitehouse, Sheldon     NVDA   2016-01-05 $1,001 - $15,000        17367.
##  3 Whitehouse, Sheldon     NVDA   2016-04-19 $1,001 - $15,000        15715.
##  4 Whitehouse, Sheldon     NVDA   2016-04-26 $1,001 - $15,000        15640.
##  5 Whitehouse, Sheldon     NVDA   2016-05-18 $1,001 - $15,000        13107.
##  6 Brenda Lulenar Lawrence NVDA   2017-03-02 $1,001 - $15,000         5618.
##  7 Debbie Dingell          NVDA   2017-03-16 $1,001 - $15,000         5346.
##  8 Whitehouse, Sheldon     NVDA   2017-01-11 $1,001 - $15,000         5261.
##  9 Pete Sessions           NVDA   2017-02-17 $1,001 - $15,000         5164.
## 10 Thomas Suozzi           NVDA   2019-01-29 $1,001 - $15,000         4188.
```

```
tail(pivot_view,10)
```

```
## # A tibble: 10 x 5
##    name                ticker traded     trade_size_usd    excess_return
##    <chr>               <chr>  <chr>      <chr>                     <dbl>
##  1 Ms. Tammy Duckworth CCL    2013-02-13 $1,001 - $15,000          -329.
##  2 Mr. Sam Farr        ACT    2013-10-01 $15,001 - $50,000         -331.
##  3 Mr. Sam Farr        ACT    2013-10-01 $15,001 - $50,000         -331.
##  4 Mr. Lamar Smith     FOSL   2014-02-10 $1,001 - $15,000          -333.
```

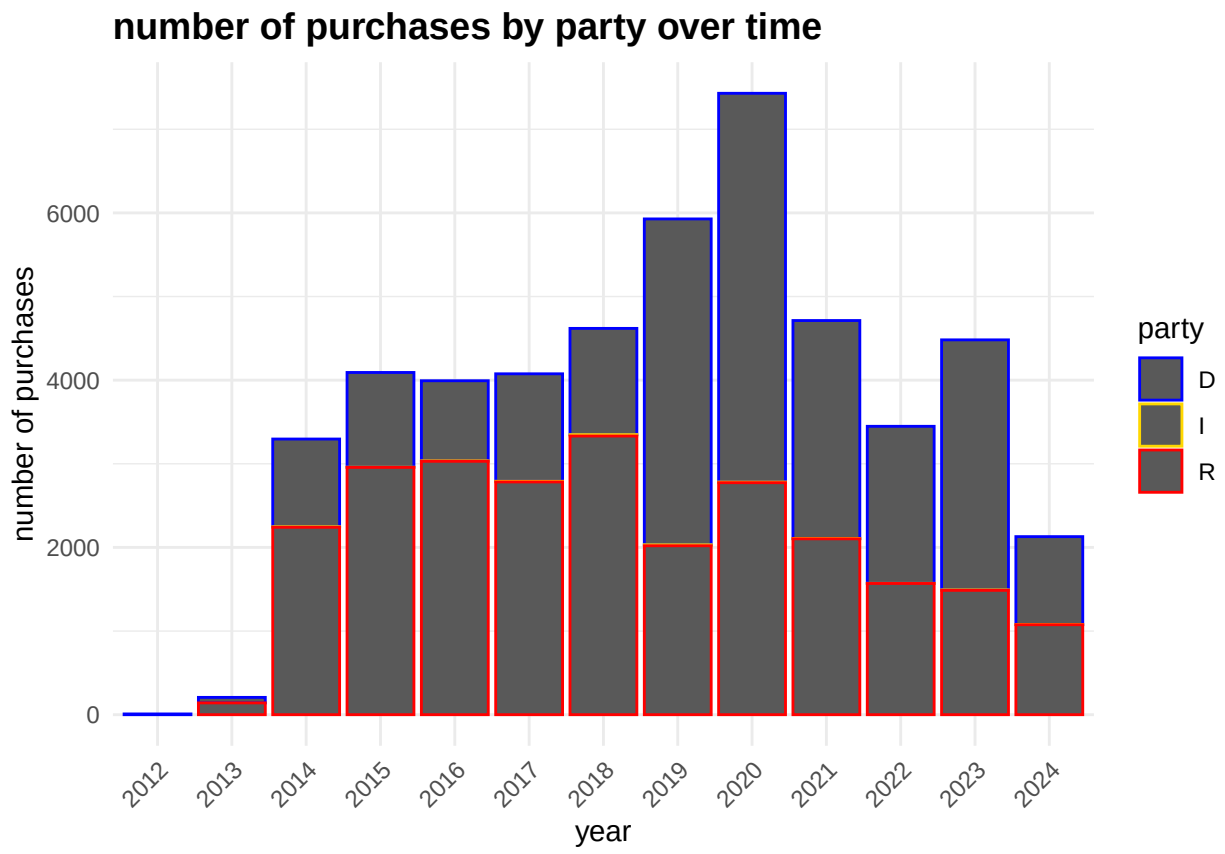
```
##  5 Mr. Beto O'Rourke   KMI    2013-01-15 $1,001 - $15,000          -335.
##  6 Mr. Lamar Smith     SWI    2013-05-20 $1,001 - $15,000          -346.
##  7 Mr. Beto O'Rourke   PAA    2013-03-28 $1,001 - $15,000          -353.
##  8 Carper, Thomas R.   DD     2012-12-17 $1,001 - $15,000          -355.
##  9 Carper, Thomas R.   DD     2012-09-13 $1,001 - $15,000          -356.
## 10 Ms. Tammy Duckworth UIS    2013-02-20 $1,001 - $15,000          -363.
```

```r
library(ggplot2)
cog$traded <- as.Date(cog$traded, format = "%Y-%m-%d")
cog$year <- format(cog$traded, "%Y")

# Filter out rows with missing 'party' or 'year'
cog_clean <- cog[!is.na(cog$party) & !is.na(cog$year), ]

# Group by year and party to count purchases
purchase_counts <- cog_clean %>%
  group_by(year, party) %>%
  summarise(purchases = n(), .groups = "drop")

# Plot the number of purchases by party over time
ggplot(purchase_counts, aes(x = year, y = purchases, color = party, group = party)) +
  geom_col() +
  theme_minimal() +
  scale_color_manual(values = c("R" = "red", "D" = "blue", "I" = "gold")) +
  labs(
    title = "number of purchases by party over time",
    x = "year",
    y = "number of purchases",
    color = "party"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(size = 14, face = "bold")
  )
```



Identifying politicians who made suspicious trades before covid

```
cov_trades <- cog %>%
  filter(traded >= as.Date("2019-12-30") & traded <= as.Date("2020-03-31"))

pivot_vov <- cov_trades[, c("name", "ticker","transaction", "traded", "trade_size_usd", "excess_return"]

pivot_vov
```

```
## # A tibble: 2,528 x 6
##    name             ticker transaction traded     trade_size_usd   excess_return
##    <chr>            <chr>  <chr>       <date>     <chr>                    <dbl>
##  1 Daniel Meuser    MSFT   Sale        2020-03-31 $1,001 - $15,000          32.6
##  2 Gilbert Cisneros DSGX   Purchase    2020-03-31 $1,001 - $15,000         104.
##  3 Gilbert Cisneros ICUI   Purchase    2020-03-31 $1,001 - $15,000        -144.
##  4 John A. Yarmuth  COP    Purchase    2020-03-31 $1,001 - $15,000         131.
##  5 Tom O'Halleran   NEM    Sale        2020-03-31 $1,001 - $15,000        -139.
##  6 Daniel Meuser    NUE    Sale        2020-03-31 $1,001 - $15,000         209.
##  7 Gilbert Cisneros ALGT   Sale        2020-03-31 $1,001 - $15,000        -141.
##  8 Greg Gianforte   UNF    Sale        2020-03-31 $50,001 - $100,~         -97.6
##  9 Daniel Meuser    CB     Sale        2020-03-31 $1,001 - $15,000          18.9
## 10 Gilbert Cisneros GBCI   Purchase    2020-03-31 $1,001 - $15,000         -61.1
## # i 2,518 more rows
```

Identifying politicians who made suspicious trades in medical stocks before covid

```
covid_stocks <- c("PFE", "MRNA", "BNTX","LLY", "JNJ", "MRK", "ABT", "TDOC", "GILD", "ISRG", "DGX", "TMO
pivot_vov2 <- cog %>%
  filter(traded >= as.Date("2019-12-30") & traded <= as.Date("2020-03-31"), ticker %in% covid_stocks)

pivot_vov3 <- pivot_vov2[, c("name", "ticker","transaction", "traded", "trade_size_usd", "excess_return

pivot_vov3
```

```
## # A tibble: 70 x 6
##    name              ticker transaction  traded     trade_size_usd excess_return
##    <chr>             <chr>  <chr>        <date>     <chr>                   <dbl>
##  1 Perdue, David     MRK    Sale (Parti~ 2020-03-31 $15,001 - $50~         -101.
##  2 Robert J. Wittman MRK    Purchase     2020-03-30 $1,001 - $15,~          -98.1
##  3 John Curtis       ISRG   Purchase     2020-03-27 $1,001 - $15,~           91.4
##  4 John Curtis       MRK    Purchase     2020-03-27 $1,001 - $15,~          -95.9
##  5 John Curtis       ABT    Purchase     2020-03-27 $1,001 - $15,~          -79.7
##  6 John Curtis       UNH    Purchase     2020-03-27 $1,001 - $15,~           21.5
##  7 John Curtis       TMO    Purchase     2020-03-27 $1,001 - $15,~          -37.0
##  8 Greg Gianforte    TDOC   Sale         2020-03-26 $50,001 - $10~         -223.
##  9 Steve Cohen       LLY    Sale         2020-03-26 $1,001 - $15,~          391.
## 10 Zoe Lofgren       JNJ    Sale         2020-03-23 $1,001 - $15,~         -130.
## # i 60 more rows
```

Plot 1-Year Performance of Politician-Traded COVID Stocks

```
library(dplyr)
library(quantmod)
```

```
## Loading required package: xts
```

```
## Loading required package: zoo
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
##
## ############################ Warning from 'xts' package ##########################
## #                                                                                #
## # The dplyr lag() function breaks how base R's lag() function is supposed to  #
## # work, which breaks lag(my_xts). Calls to lag(my_xts) that you type or       #
## # source() into this session won't work correctly.                               #
## #                                                                                #
## # Use stats::lag() to make sure you're not using dplyr::lag(), or you can add #
```

```
## # conflictRules('dplyr', exclude = 'lag') to your .Rprofile to stop           #
## # dplyr from breaking base R's lag() function.                                #
## #                                                                             #
## # Code in packages is not affected. It's protected by R's namespace mechanism #
## # Set `options(xts.warn_dplyr_breaks_lag = FALSE)` to suppress this warning.  #
## #                                                                             #
## ###############################################################################

##
## Attaching package: 'xts'

## The following objects are masked from 'package:dplyr':
##
##     first, last

## Loading required package: TTR

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(ggplot2)

# Step 1: Filter COVID-related purchase trades only
pivot_purchases <- cog %>%
  filter(ticker %in% covid_stocks,
         traded >= as.Date("2019-12-30"),
         traded <= as.Date("2020-03-31"),
         transaction == "Purchase") %>%
  select(name, ticker, transaction, traded, trade_size_usd, excess_return)

# Step 2: Unique tickers politicians purchased
covid_purchase_tickers <- unique(pivot_purchases$ticker)

# Step 3: Pull stock price data
get_stock_data <- function(ticker) {
  df <- getSymbols(ticker, src = "yahoo", from = "2020-03-01", to = "2021-03-01", auto.assign = FALSE)
  data.frame(
    date = index(df),
    price = as.numeric(Cl(df)),
    ticker = ticker
  )
}

price_data <- do.call(rbind, lapply(covid_purchase_tickers, get_stock_data))

# Step 4: Calculate percent growth
percent_growth <- price_data %>%
  group_by(ticker) %>%
  mutate(percent_change = (price - first(price)) / first(price) * 100)

# Step 5: Count purchases per ticker
trade_counts <- pivot_purchases %>%
```

```r
  group_by(ticker) %>%
  summarise(trade_count = n(), .groups = "drop")

# Step 6: Merge and relabel
percent_growth_labeled <- percent_growth %>%
  left_join(trade_counts, by = "ticker") %>%
  mutate(ticker_label = paste0(ticker, " (", trade_count, " trades)"))

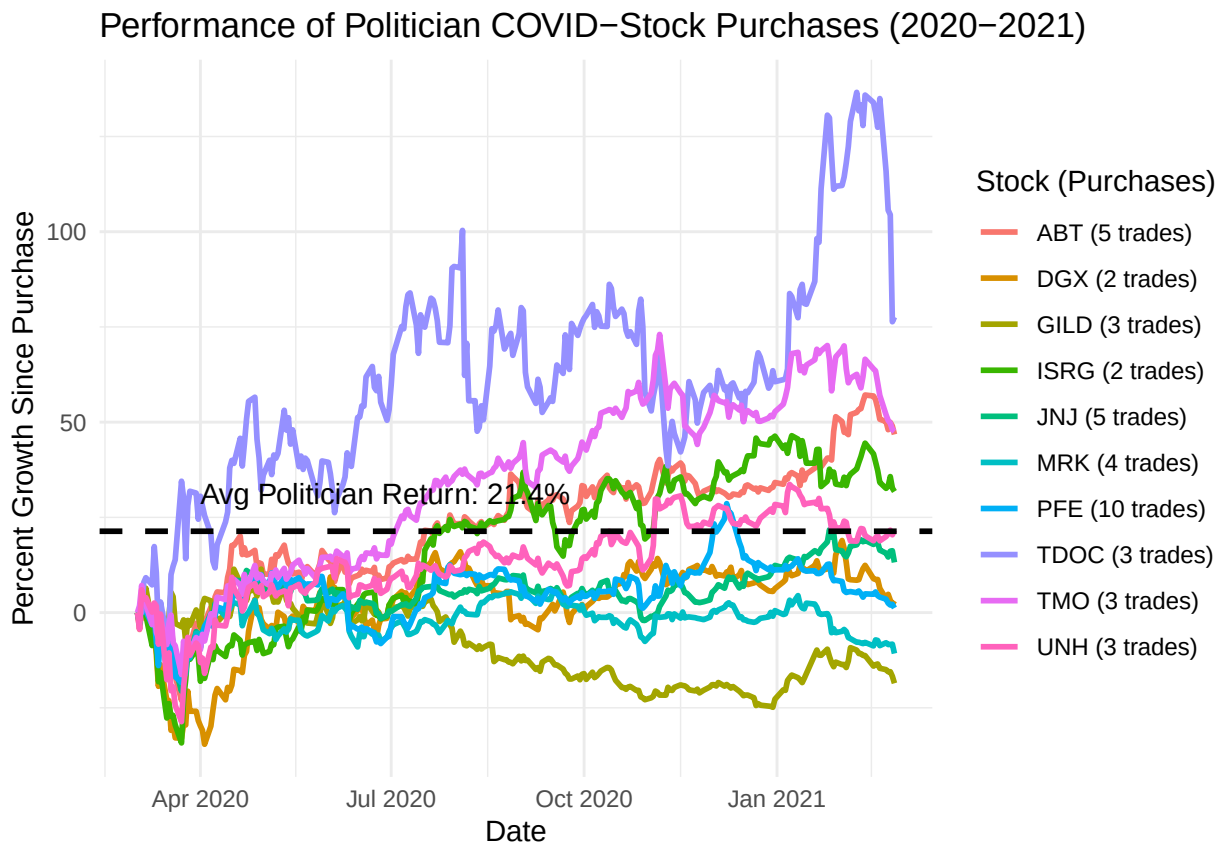
# Step 7: Calculate average return
final_returns <- percent_growth_labeled %>%
  group_by(ticker_label) %>%
  filter(date == max(date)) %>%
  summarise(final_return = mean(percent_change))

avg_return <- mean(final_returns$final_return)

# Step 8: S&P 500 return
sp500 <- getSymbols("^GSPC", src = "yahoo", from = "2020-03-01", to = "2021-03-01", auto.assign = FALSE)
sp500_ret <- (last(Cl(sp500)) - first(Cl(sp500))) / first(Cl(sp500)) * 100

# Step 9: Final plot
ggplot(percent_growth_labeled, aes(x = date, y = percent_change, color = ticker_label)) +
  geom_line(size = 1) +
  geom_hline(yintercept = avg_return, linetype = "dashed", color = "black", size = 1) +
  annotate("text", x = as.Date("2020-04-01"), y = avg_return + 10,
           label = paste0("Avg Politician Return: ", round(avg_return, 1), "%"),
           hjust = 0, size = 4, color = "black") +
  geom_hline(yintercept = sp500_ret, linetype = "dotted", color = "gray40", size = 1) +
  annotate("text", x = as.Date("2020-04-01"), y = sp500_ret - 10,
           label = paste0("S&P 500 Return: ", round(sp500_ret, 1), "%"),
           hjust = 0, size = 4, color = "gray40") +
  labs(
    title = "Performance of Politician COVID-Stock Purchases (2020-2021)",
    x = "Date",
    y = "Percent Growth Since Purchase",
    color = "Stock (Purchases)"
  ) +
  theme_minimal()
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```




```r
library(dplyr)
library(quantmod)
library(ggplot2)

# Step 1: Filter all purchases in early COVID period
pivot_top <- cog %>%
  filter(traded >= as.Date("2019-12-30"),
         traded <= as.Date("2020-03-31"),
         transaction == "Purchase") %>%
  select(name, ticker, transaction, traded, trade_size_usd, excess_return)

# Step 2: Find top 20 most purchased tickers
top_tickers <- pivot_top %>%
  group_by(ticker) %>%
  summarise(trade_count = n(), .groups = "drop") %>%
  arrange(desc(trade_count)) %>%
  slice_head(n = 10)

# Step 3: Get price data for top 20 tickers
get_stock_data <- function(ticker) {
  tryCatch({
    df <- getSymbols(ticker, src = "yahoo", from = "2020-03-01", to = "2021-03-01", auto.assign = FALSE)
    data.frame(
      date = index(df),
      price = as.numeric(Cl(df)),
      ticker = ticker
```

```
    )
  }, error = function(e) {
    NULL
  })
}

price_data <- do.call(rbind, lapply(top_tickers$ticker, get_stock_data))

# Step 4: % growth over time
percent_growth <- price_data %>%
  group_by(ticker) %>%
  mutate(percent_change = (price - first(price)) / first(price) * 100)

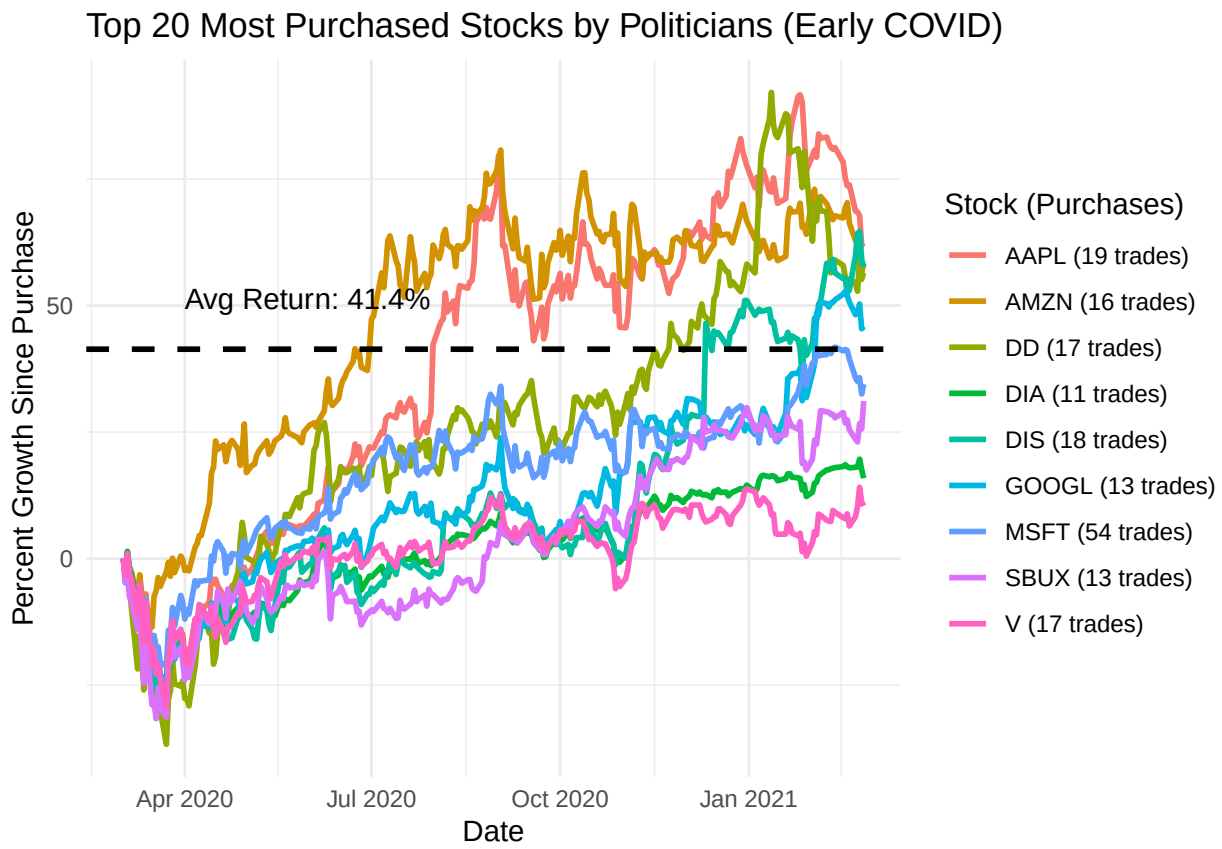
# Step 5: Add labels with purchase counts
percent_growth_labeled <- percent_growth %>%
  left_join(top_tickers, by = "ticker") %>%
  mutate(ticker_label = paste0(ticker, " (", trade_count, " trades)"))

# Step 6: Avg return across top 20
final_returns <- percent_growth_labeled %>%
  group_by(ticker_label) %>%
  filter(date == max(date)) %>%
  summarise(final_return = mean(percent_change))

avg_return <- mean(final_returns$final_return)

# Step 7: S&P 500 return
sp500 <- getSymbols("^GSPC", src = "yahoo", from = "2020-03-01", to = "2021-03-01", auto.assign = FALSE)
sp500_ret <- (last(Cl(sp500)) - first(Cl(sp500))) / first(Cl(sp500)) * 100

# Step 8: Plot
ggplot(percent_growth_labeled, aes(x = date, y = percent_change, color = ticker_label)) +
  geom_line(size = 1) +
  geom_hline(yintercept = avg_return, linetype = "dashed", color = "black", size = 1) +
  annotate("text", x = as.Date("2020-04-01"), y = avg_return + 10,
           label = paste0("Avg Return: ", round(avg_return, 1), "%"),
           hjust = 0, size = 4, color = "black") +
  geom_hline(yintercept = sp500_ret, linetype = "dotted", color = "gray40", size = 1) +
  annotate("text", x = as.Date("2020-04-01"), y = sp500_ret - 10,
           label = paste0("S&P 500 Return: ", round(sp500_ret, 1), "%"),
           hjust = 0, size = 4, color = "gray40") +
  labs(
    title = "Top 20 Most Purchased Stocks by Politicians (Early COVID)",
    x = "Date",
    y = "Percent Growth Since Purchase",
    color = "Stock (Purchases)"
  ) +
  theme_minimal()
```



NOTABLE stock, ($DD) DuPont, a company making personal protective equipment

```
library(dplyr)
library(ggplot2)

# Step 1: Filter early COVID trades
heatmap_data <- cog %>%
  filter(traded >= as.Date("2019-12-30"),
         traded <= as.Date("2020-03-31"),
         !is.na(excess_return))

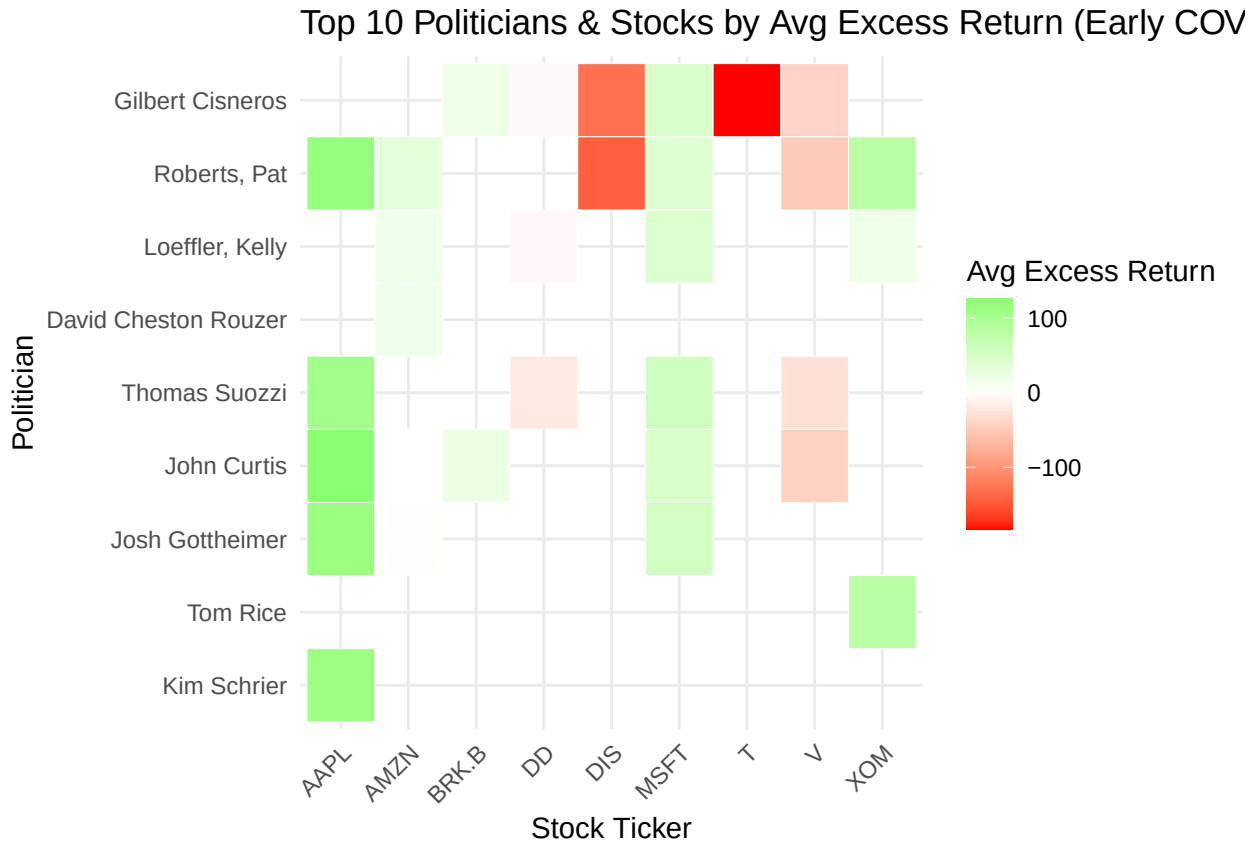
# Step 2: Find top 10 politicians by total excess_return
top_names <- heatmap_data %>%
  group_by(name) %>%
  summarise(total_return = sum(excess_return)) %>%
  arrange(desc(total_return)) %>%
  slice_head(n = 10) %>%
  pull(name)

# Step 3: Find top 10 tickers by trade count
top_tickers <- heatmap_data %>%
  count(ticker) %>%
  arrange(desc(n)) %>%
  slice_head(n = 10) %>%
  pull(ticker)

# Step 4: Filter to those combos and get avg return
```

```r
heatmap_filtered <- heatmap_data %>%
  filter(name %in% top_names, ticker %in% top_tickers) %>%
  group_by(name, ticker) %>%
  summarise(avg_return = mean(excess_return), .groups = "drop")

# Step 5: Plot clean heatmap
ggplot(heatmap_filtered, aes(x = ticker, y = reorder(name, -avg_return), fill = avg_return)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "red", high = "green", mid = "white", midpoint = 0,
                       name = "Avg Excess Return") +
  labs(
    title = "Top 10 Politicians & Stocks by Avg Excess Return (Early COVID Trades)",
    x = "Stock Ticker",
    y = "Politician"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Gilbert Cisneros: Accused of making extensive stock trades following a private congressional briefing on the pandemic, reportedly trading up to $6.8 million worth of stock. His transactions were noted as among the most extensive by any member of Congress during this period.

Kelly Loeffler: Accused of selling stocks after attending a private Senate briefing on COVID-19 in January 2020. Her actions drew significant scrutiny, although investigations by the Justice Department and Senate Ethics Committee were eventually closed without charges.

John Curtis: Allegedly made well-timed trades, including purchasing Abbott Laboratories stock on the same day it was awarded a federal grant in March 2020. While these trades raised ethical concerns, they did not violate any laws.

Kim Schrier: Identified in an analysis as one of several U.S. representatives whose transactional activity during the pandemic suggested potential insider trading.

Tom Rice: Also flagged in analyses for possible insider trading based on financial disclosures during the COVID-19 period.